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graph LR; Start(( )) --> S1((S1)); S1 --> S1; S1 --> S2; S2 --> S1; S2 --> S2;
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State transition diagram for a finite state automaton with two states,  $S_1$  and  $S_2$ .

States:

- $S_1$  (Start state)
- $S_2$  (Final state)

Transitions:

- $S_1 \rightarrow S_1$ :  $(1E, LAT. AQ) / (LAT. AQ, NUCC)$
- $S_1 \rightarrow S_2$ :  $(1E, LAT. COLA) / (NUCC, NUCC)$
- $S_2 \rightarrow S_1$ :  $(1E, LAT. COLA) / (LAT. COLA, NUCC)$
- $S_2 \rightarrow S_2$ :  $(2E, LAT. COLA) / (LAT. COLA, DEST. 1E)$
- $S_1 \rightarrow S_1$  (bottom loop):  $(2E, LAT. COLA) / (LAT. COLA, NUCC)$
- $S_1 \rightarrow S_1$  (left loop):  $(2E, LAT. AQ) / (LAT. AQ, DEST. 1E)$